Module Overview

Essential Question: How do people and animals use their senses to navigate the world?
Quote: "All our knowledge begins with the senses." — Immanuel Kant
Big Idea Words (4): perception, tactile, aroma, distinguish
Main Selections (workbook 4):

- The Science Behind Sight · Louise Spilsbury · Informational Text
- Animal Senses · from Animal Atlas · Informational Video (Media)
- Blind Ambition · Matthew Cooper, as told to Rachel Buchholz (photos by Karine Aigner) · Personal Narrative
- The Game of Silence · Louise Erdrich (illus. Paolo Domeniconi) · Historical Fiction

Note. The SoT lists 26 core vocabulary items on Worksheet A (4 Big Idea + 5 The Science Behind Sight + 4 Animal Senses + 6 Blind Ambition + 7 The Game of Silence), in myBook appearance order. This G4 Teacher's Guide labels each selection's panel "Critical Vocabulary" — the string "Power Words" does not appear in the source. The teacher "Words for Using Context Clues" / "Multiple-Meaning Words" strategy boxes (opaque, pupil, relish/deal, blind, club, snatch, serene, field, downside, team) are NOT counted as core vocabulary. "What Are the Five Senses?" is the Big-Idea Infographic Short Read (no Critical Vocabulary panel) and carries the 4 Big Idea Words; it is excluded from the per-selection workbook units. This module DOES include one Media selection — the "Animal Senses" Informational Video — which has its own Critical Vocabulary (4 words). It is separate from the module-opener "Get Curious" video "Super Senses."

Answer Distribution (78 Multiple-Choice Items)

Verified counts across Worksheets A (26), B (32), and D (20). Worksheet C is open-response IB and not counted in MCQ balance.

A · 20 · B · 20 · C · 19 · D · 19

These four totals are the count of correct-answer LETTERS (A/B/C/D) across all 78 multiple-choice items — 20 answers are "A", 20 are "B", 19 are "C", 19 are "D" (as even as 78 allows). Balance enforced via choice-order rotation only; correct-answer meanings and option texts are preserved from the Source of Truth. All 26 SoT vocabulary words covered on Worksheet A in myBook appearance order (Big Idea → The Science Behind Sight → Animal Senses → Blind Ambition → The Game of Silence).

Voca Trap Items (Worksheet A)

7 of 26 ≈ 27%, all flagged below. Trap pattern: polysemy / sound-alike / opposite-meaning baits drawn from the natural meaning field of each word — no hallucination.

- Q2 **tactile** — sense-confusion bait ("done quietly," "seen only in bright light") vs. "experienced through the sense of touch"
- Q7 **reflect** — polysemy bait ("thinks quietly about a problem") vs. "bounces back without passing through the surface"
- Q9 **judge** — polysemy bait (courtroom "decide who is guilty / say someone did wrong") vs. "guess or estimate size or amount"
- Q10 **relish** — homograph bait (the condiment "put it on a hot dog") vs. "enjoy it very much"
- Q16 **command** — opposite bait ("by yourself, with no one else") vs. "because you were told to"
- Q20 **absurd** — tone bait ("sad and serious," "hard and tiring") vs. "silly or ridiculous"
- Q23 **despised** — weak-feeling bait ("felt unsure," "were curious") vs. "felt a strong dislike for it"

Reading Trap Items (Worksheet B)

3 of 32 ≈ 9%, all flagged below. Trap pattern for Section B: THEME/MAIN-idea bait (a true detail mistaken for the central idea) and length-balanced correct answers so a longest-option-wins guess fails. Distractors built from SoT only — no hallucination. Three anchor selections are INFORMATIONAL (RI: The Science Behind Sight, Animal Senses video, Blind Ambition); one is LITERARY (RL: The Game of Silence). Codes used are RI.4.1 / .2 / .3 / .5 / .6 / .8 and RL.4.1 / .2 / .3 only. **RI.4.4, RI.4.7, and RL.4.7 are NOT used**, and there are no "from the text" snippet boxes.

- Q16 **Animal Senses main idea** — "a rhinoceros has weak eyesight," "some animals hunt at night," "a narrator speaks" are true details, baited against the real central idea: animals depend on their senses, each in its own way, to survive
- Q24 **Blind Ambition central idea** — "got Twyla a year and a half ago," "runs the sound system," "reads Braille and tells jokes" bait vs. by accepting and adapting, a person can use other senses and help to live a full, capable life
- Q32 **The Game of Silence theme** — "burning wand by the ankle," "Old Tallow smelling of wolf," "Nokomis sang four times" bait vs. silence and careful attention help us understand what matters most, such as home

Source-of-Truth Verification

Into Reading · Grade 4 · Module 2 · For Teacher Use

VERIFIED

How Every Item Is Grounded

Every item on this Answer Key traces to the Source-of-Truth JSON for G4 Module 2, built from the Teaching Pal OCR (3,675 lines, 33 spread-pages) and verified against the PDF intro spreads (every Critical Vocabulary panel was read from a high-resolution crop). No characters, words, or events outside the Source of Truth were added.

- **Vocabulary (Worksheet A):** all 26 words come from the module's Big Idea Words (4: perception, tactile, aroma, distinguish) and the four "Critical Vocabulary" panels — The Science Behind Sight 5, Animal Senses 4, Blind Ambition 6, The Game of Silence 7. Each meaning is checked against the myBook inline footnote definition. The teacher "Words for Using Context Clues" / "Multiple-Meaning Words" strategy boxes (opaque, pupil, relish/deal, blind, club, snatch, serene, field, downside, team) are deliberately NOT counted as core vocabulary.
- **Terminology:** this G4 Teacher's Guide labels each panel "Critical Vocabulary," not "Power Words" — the workbook uses that exact label, matching the source.
- **Reading (Worksheet B):** the 32 items use RI.4 codes for the three informational anchors (The Science Behind Sight, the Animal Senses video, Blind Ambition) and RL.4 codes for the one literary anchor (The Game of Silence). Each correct answer is grounded in the Source of Truth; distractors are SoT-anchored, never invented. RI.4.4, RI.4.7, and RL.4.7 are not used, and there are no "from the text" snippet boxes.
- **Genre & credits:** The Science Behind Sight = Informational Text (Louise Spilsbury); Animal Senses = Informational Video (from Animal Atlas, narrator — no individual byline); Blind Ambition = Personal Narrative (Matthew Cooper as told to Rachel Buchholz, photographs by Karine Aigner); The Game of Silence = Historical Fiction (Louise Erdrich, illus. Paolo Domeniconi).
- **IB (Worksheet C):** the PYP concepts — Function, Perspective, Responsibility, Causation, Connection — follow the Source-of-Truth senses focus of this module. The Essential Question text on Q5 matches the workbook word for word. Korean teacher guidance appears in Section C only; Sections A, B, and D are in English.
- **MAP (Worksheet D):** four NEW passages (167 / 172 / 172 / 184 words), RIT 200→220, Lexile 500L–800L. None recall the four anchor selections; each applies the EQ to a fresh way the senses help a creature or person navigate the world.
- **Answer balance:** 78 multiple-choice items distribute as 20 "A" / 20 "B" / 19 "C" / 19 "D" (as even as 78 allows), achieved by choice-order rotation only — every meaning is preserved from the Source of Truth.

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Owl's Night Eyes · **167 words** · Lexile 500–560L · RIT 200–206
- Passage 2 · How Dogs Smell the World · **172 words** · Lexile 560–620L · RIT 206–211
- Passage 3 · Reading with Your Fingers · **172 words** · Lexile 620–700L · RIT 211–216
- Passage 4 · The Taste Test · **184 words** · Lexile 700–800L · RIT 216–220

Word counts match the workbook passage meta exactly. All four within target band 150–220 words; RIT 200→220; Lexile 500L–800L. Each passage models a different way the senses help navigate the world — an owl's night vision and hearing, a dog's sense of smell, the sense of touch through braille, and how taste and smell work together. Passages are NEW — no recall of the four anchor selections.

What This Module Does NOT Contain (Guardrails)

- No "Power Words" label — the G4 source says "Critical Vocabulary."
- No Worksheet E — this workbook series uses four worksheets (A · B · C · D).
- The module quote IS by a NAMED person, Immanuel Kant, and is the EPIGRAPH only — he is NOT a story author.

- "What Are the Five Senses?" is the Big-Idea Infographic Short Read (no Critical Vocabulary panel) and is excluded from the per-selection units; it carries the 4 Big Idea Words instead. Its author is unverified — no byline in the source.
- The "Animal Senses" Informational Video IS a numbered body selection with its own Critical Vocabulary (4 words). It is separate from the module-opener "Get Curious" intro video "Super Senses."
- Critical Vocabulary counts are 5 / 4 / 6 / 7 = 22, not padded to match. Big Idea (4) + Critical (22) = 26 total.
- "The Game of Silence" is set in an 1850 Ojibwe community on Lake Superior. "Pinch" is Omakayas's younger brother; "Two Strike" and "the Angry One" are separate cousins/visitors — do not conflate them with Pinch.
- Module 3 "Rise to the Occasion" (Never Give Up!, Rent Party Jazz, Hurricanes, My Diary from Here to There, etc.) is a DIFFERENT module and is NOT included here.
- Performance Task = "Write an Expository Essay." It is not a narrative or opinion essay.

Worksheet A — Vocabulary · Answer Key

26 items · CCSS L.4.4 / L.4.5 / L.4.6

BIG IDEA WORDS

Q1 L.4.6

perception

C

Meaning: your perception of something is how you notice or experience it using your senses. **Source:** SoT Big Idea Word — Vocabulary Card 2.1. *Distractors A/B/D give a drawing / a fixed fact / a name.*

Q2 L.4.5 TRAP

tactile

A

Meaning: something tactile is experienced through the sense of touch. **Source:** SoT Big Idea Word 2.2. **Trap:** B/D bait other senses (quiet = hearing, bright light = sight).

Q3 L.4.6

aroma

B

Meaning: an aroma is a strong, pleasant smell. **Source:** SoT Big Idea Word 2.3. *Distractors A/C/D give a noise / a color / a texture (all the wrong sense).*

Q4 L.4.6

distinguish

D

Meaning: if you distinguish things, you can tell how they are different from one another. **Source:** SoT Big Idea Word 2.4. *Distractors give mix / hide / trade.*

CRITICAL VOCABULARY · THE SCIENCE BEHIND SIGHT

Q5 L.4.4

luminous

A

Evidence: SoT — "If something is luminous, it gives off light" (OCR line 431); "The sun is luminous." *Distractors give floats / loud / cold.*

Q6 L.4.4

transparent

C

Evidence: SoT — "If an object is transparent, you can see through it" (OCR line 432). *Distractors give hear through / bend / too heavy.*

Q7 L.4.5 TRAP

reflect

B

Evidence: SoT — "When light reflects off a surface, it bounces back without passing through the surface" (OCR lines 546–547). **Trap:** C baits the other sense of reflect ("think quietly").

Q8 L.4.4

illuminates

D

Evidence: SoT — "Something that illuminates gives off light and makes the area around it brighter" (OCR lines 549–550). *Distractors give cools / quiets / cleans.*

Q9 L.4.5 TRAP

judge

A

Evidence: SoT — "If you judge, you guess or estimate the size or amount of something" (OCR lines 752–753); two eyes help judge distance. **Trap:** B/C/D bait the courtroom sense of "judge."

CRITICAL VOCABULARY · ANIMAL SENSES

Q10 L.4.5 TRAP

relish

C

Evidence: SoT — "If you relish something, you enjoy it very much" (OCR line 1577). **Trap:** A baits the homograph "relish" (the condiment).

Q11 L.4.4

familiar

D

Evidence: SoT — "Something familiar is something that you know and are used to" (OCR line 1578). *Distractors give afraid of / never met / only at night.*

Q12 L.4.4

savor

A

Evidence: SoT — "If you savor something, you take your time enjoying it" (OCR line 1580). *Distractors give swallow fast / throw away / give away.*

Q13 L.4.4

enhance

B

Evidence: SoT — "If you enhance something, you make it into something better or improve upon it" (OCR lines 1586–1587). *Distractors give hide / count / break apart.*

CRITICAL VOCABULARY · BLIND AMBITION

Q14 L.4.4

accepted

C

Evidence: SoT — "If you have accepted a situation, you understand that it can't be changed" (OCR line 1864). *Distractors give still changing / forgotten / never noticed.*

Q15 L.4.4

obstacles

D

Evidence: SoT — "Obstacles are objects that make it hard to get where you want to go" (OCR line 1865); Twyla helps Matthew get around them. *Distractors give friends / sounds / directions.*

Q16 L.4.5 TRAP

command

A

Evidence: SoT — "If you do something on command, you do it because you were told to" (OCR line 1864). **Trap:** B baits the opposite ("by yourself, with no one else").

Q17 L.4.4

denying

B

Evidence: SoT — "Denying something means not believing that it's true" (OCR line 1865). *Distractors give explaining / writing down / asking for more.*

Q18 L.4.4

adapt

C

Evidence: SoT — "If you adapt to something, you figure out how to deal with it" (OCR line 1985); "To fit in, I know that I have to adapt." *Distractors give refuse / run away / wait it out.*

Q19 L.4.4

comfort

D

Evidence: SoT — "If you comfort someone, you say or do things to make the person feel better" (OCR line 2162). *Distractors give leave alone / tell wrong / make worse.*

Q20 L.4.5 TRAP

absurd

A

Evidence: SoT — "If something is absurd, you think it is silly or ridiculous" (OCR line 2534). **Trap:** B/C bait the wrong tone ("sad and serious," "hard and tiring").

Q21 L.4.4

taunt

B

Evidence: SoT — "A taunt is something someone says to anger or upset someone else" (OCR line 2763); Pinch taunts Omakayas. *Distractors give a gift / a song / a question.*

Q22 L.4.4

forfeit

C

Evidence: SoT — "If you forfeit something, you lose it because you have broken a rule" (OCR line 2778). *Distractors give win / share / hide.*

Q23 L.4.5 TRAP

despised

D

Evidence: SoT — "If you despised something, you felt a strong dislike for it" (OCR line 2898). **Trap:** A/B/C bait weak feelings ("unsure," "curious") instead of strong dislike.

Q24 L.4.4

ferocious

A

Evidence: SoT — "Something that is ferocious is very fierce, mean, and violent" (OCR line 3085); Old Tallow's ferocious dogs. *Distractors give gentle / sleepy / shy.*

Q25 L.4.4

elaborately

B

Evidence: SoT — "If something is elaborately dressed or decorated, it has many complex artistic details" (OCR lines 3113–3114). *Distractors give no decoration / one plain color / a smell.*

Q26 L.4.4

coveted

C

Evidence: SoT — "If you coveted something, you wanted it very much" (OCR line 3199); Pinch coveted the prize. *Distractors give already owned / gave away / afraid of.*

Worksheet B — Reading Comprehension · Answer Key

32 items · CCSS RI.4.1 / .2 / .3 / .5 / .6 / .8 · RL.4.1 / .2 / .3

Selection 1 · The Science Behind Sight

INFORMATIONAL TEXT · 8 ITEMS

Q1 RI.4.2

B

Central Idea: the text explains how light and the eye work together so people and animals can see (SoT key text, OCR lines 333–335, 527–532, 643–652). *Others pick a stray detail.*

Q2 RI.4.1

C

Evidence: "Things that give off light are luminous. The sun is luminous" (OCR lines 412–415). *D is the definition of a reflective object, not luminous.*

Q3 RI.4.3

A

Evidence: most things are not luminous; "they reflect it! Light bounces off surfaces... when some of that reflected light enters our eyes" (OCR lines 527–532). *Others off-text.*

Q4 RI.4.3

D

Evidence: "In bright light... the pupil becomes smaller. In low light, the pupil becomes" larger (OCR lines 577–579). *Others reverse or off-text.*

Q5 RI.4.3

C

Evidence: "The image that the lens sends to the retina is upside down. The brain turns it the right way up!" (OCR lines 650–652). *Others off-text.*

Q6 RI.4.8

A

Reasons & Evidence: "Foxes, lions, and most other predators... have two eyes on the front of their head... This helps them to hunt and catch animals" (OCR lines 761–763). *Others off-text.*

Q7 RI.4.3

D

Evidence: "A cat's eyes glow in the dark because they have a part behind their retinas that works a bit like a mirror. It reflects light back onto the retina" (OCR lines 865–867). *Others off-text.*

Q8 RI.4.5

B

Text Features: SoT — headings, labeled diagrams, and captions organize the information and show how parts of the eye work (SoT text-feature note). *Others off-text.*

Selection 2 · Animal Senses

INFORMATIONAL VIDEO · 8 ITEMS

Q9 RI.4.2

D

Central Idea: the video shows how different animals use their senses to find food/prey and stay safe from predators (OCR lines 1538–1548). *Others off-topic.*

Q10 RI.4.6

B

Evidence: "A narrator explains what is happening on the screen" (OCR line 1368). *Others off-text.*

Q11 RI.4.6

A

Evidence: "nature is the great compensator" — nature will compensate or make up for one sense that isn't as strong (OCR lines 1536–1571). *Others off-text.*

Q12 RI.4.3

C

Evidence: a rhinoceros "does not have good eyesight, but it isn't necessary because the rhino has no natural predators" (OCR lines 1568–1571). *Others off-text.*

Q13 RI.4.3

B

Evidence: animals "use touch to find their way around their habitats" (OCR lines 1538–1548). *Others off-text.*

Q14 RI.4.2

D

Media: SoT — a video uses a narrator together with moving pictures and sound; "information... supported through the visuals" (OCR lines 1368, 1490–1497). *Others off-text.*

Q15 RI.4.3

A

Evidence: "You can tell if an animal is a predator or a prey animal based on how" it uses its senses to hunt or escape (OCR lines 1581–1583). *Others off-text.*

Q16 RI.4.2 TRAP

C

Main Idea: animals depend on their senses, each in its own way, to survive (OCR lines 1538–1548). **Trap:** A/B/D are true details, not the main idea.

Selection 3 · Blind Ambition

PERSONAL NARRATIVE · 8 ITEMS

Q17 RI.4.2

C

Central Idea: Matthew, who is blind, adapts and uses his other senses, with help from Twyla, to do what he loves (OCR lines 1825–1833, 1943–1972). *Others off-topic.*

Q18 RI.4.3

A

Evidence: "When I was three I was diagnosed with a brain tumor that press[ed] against my optic nerve, and I gradually lost my sight" (OCR lines 1825–1829). *Others off-text.*

Q19 RI.4.3

D

Evidence: Twyla "helps me cross the street... She helps him cross streets, get around obstacles, and walk through crowded hallways at school" (OCR lines 1852–1857). *Others off-text.*

Q20 RI.4.3

B

Evidence: "I make them wear buzzers so I know where they are... I put Braille labels on all the controls" (OCR lines 1957–1963). *Others off-text.*

Q21 RI.4.6

D

Point of View: SoT — Matthew narrates in the first person ("I am Matthew Cooper"), so the reader hears his own thoughts and feelings from the inside (OCR lines 1840–1846). *Others misread it.*

Q22 RI.4.3

A

Evidence: "my coach or my dad will position my club behind the ball and tell me how far away the hole is. Then I swing" (OCR lines 1969–1972). *Others off-text.*

Q23 RI.4.8

B

Reasons & Evidence: "There's always something life takes from you, and something life gives back. Like, I think being blind makes me a better golfer" (OCR lines 2145–2147). *Others off-text.*

Q24 RI.4.2 TRAP

C

Central Idea: by accepting his situation and adapting, Matthew uses other senses and help to live a full, capable life (SoT theme). **Trap:** A/B/D are true details, not the central idea.

Selection 4 · The Game of Silence

HISTORICAL FICTION · 8 ITEMS

Q25 RL.4.2

C

Central Idea: Omakayas, an Ojibwe girl, struggles to stay silent in the game while the adults prepare to discuss serious news (OCR lines 2479–2489). *Others off-topic.*

Q26 RL.4.3

D

Evidence: Pinch teases her, makes her doll teeter on his knees, and "drew the burning wand from the fire and laid it innocently next to her ankle" (OCR lines 2634–2636, 2750–2752). *Others off-text.*

Q27 RL.4.3

A

Evidence: "Omakayas pretended to shut her eyes again, but cleverly... she grabbed... and Omakayas triumphantly clutched her" doll (OCR lines 2641–2645). *Others off-text.*

Q28 RL.4.3

B

Evidence: the children play the silent game "while they wait for others to arrive so the adults can discuss what is happening to their people" (OCR lines 2484–2489). *Others off-text.*

Q29 RL.4.3

B

Evidence: by the end "they knew the threat of a much bigger loss... That something was home" — the playful game turns serious (OCR lines 3306–3314). *Others off-text.*

Q30 RL.4.3

D

Craft: SoT — "Nokomis's old bones ached, and she creaked like a tree every time she moved" — the simile shows she is very old (OCR lines 2625–2626). *Others misread the comparison.*

Q31 RL.4.3

A

Evidence: Pinch's taunts (the doll, the burning wand) make Omakayas hold back her anger so she will not cry out and lose (OCR lines 2503–2521, 2705–2713). *Others off-text.*

Q32 RL.4.2 TRAP

C

Theme: silence and careful attention help the children grasp what matters most — the value of home (OCR lines 3306–3314). **Trap:** A/B/D are stray details.

Worksheet C — IB Critical Thinking · Answer Key

5 questions · PYP Key Concepts · DOK 2–4

Q1 Function DOK 3

In **"The Science Behind Sight,"** show how light travels so that we can see, then explain WHY each part of the eye matters — and why we see objects that are not luminous.

MODEL ANSWER

Light Starts: Seeing begins with light. Luminous things, such as the sun or a lamp, give off their own light. That light travels in straight lines and lands on the objects around us. **Into the Eye:** Light enters the eye through a small opening called the pupil. In bright light the pupil becomes smaller to let in less light; in low light it becomes larger to let in more. The light passes through a lens, which focuses the image onto the retina at the back of the eye. **To the Brain:** The image on the retina is actually upside down. The optic nerve carries the signals from the retina to the brain, and the brain turns the picture the right way up so that we understand what we are seeing. **Why we see non-luminous objects:** Most things around us do not give off their own light. We can see them because light bounces, or *reflects*, off their surfaces and into our eyes. Shiny surfaces like mirrors reflect light best, while black fabric reflects very little. So sight is really a chain: light from a luminous source reflects off an object, enters the eye through the pupil and lens, lands on the retina, and travels through the optic nerve to the brain. Each part has its own function, and together they let us see.

KEYWORDS (3)

light = luminous source, travels straight

eye = pupil, lens, retina, optic nerve

non-luminous = seen by reflected light

SCORING NOTES

- Full credit (3): light source + path through the eye (pupil/lens/retina) + brain/optic nerve + why we see reflected (non-luminous) objects.
- Partial (2): names some eye parts but no clear light-to-brain path or no use of "reflect."
- Minimal (1): only "we see with our eyes" with no steps.

TEACHING STEPS

Step 1 · Three boxes in order.

강사: *"Label three boxes: Light Starts, Into the Eye, To the Brain. What goes in each?"*

코멘트. 상자 세 칸 — 빛의 출발·눈 속·뇌까지. 각 칸에 단계 적기.

Step 2 · The eye parts.

강사: *"Underline the eye parts in order: pupil, lens, retina, optic nerve."*

코멘트. 눈의 부분을 순서대로 밑줄 — 동공·렌즈·망막·시신경.

Step 3 · Reflect.

강사: *"Finish: 'We see a non-luminous object because light ___ off it.'"*

코멘트. 문장 완성 — '빛을 내지 않는 물체가 보이는 이유는 빛이 거기에서 ____하기 때문이다.'

STUDENT PROMPTS

- "Why does the pupil change size in bright and dim light?"
  밝을 때와 어두울 때 동공 크기가 왜 달라질까?
- "What does the brain do with the upside-down image?"
  뇌는 거꾸로 된 상을 어떻게 처리하니?
- "How is seeing a chair different from seeing the sun?"
  의자를 보는 것과 태양을 보는 것은 어떻게 다르니?

DIFFERENTIATION

STRUGGLING Put 4 cards (light / pupil / retina / brain) in order.
카드 4장 — 빛·동공·망막·뇌 순서 배열.

ADVANCED Explain why a cat sees better at night using the word "reflect."
'reflect'를 써서 고양이가 밤에 더 잘 보는 이유 설명.

COMMON MISCONCEPTIONS

PITFALL "Our eyes shine light onto things." We see because light reflects off objects INTO our eyes.
눈이 빛을 쏘는 게 아님 — 물체에서 반사된 빛이 눈으로 들어옴.

PITFALL "Every object is luminous." Most objects only reflect light; few give off their own.
모든 물체가 발광체는 아님 — 대부분은 빛을 반사만 함.

LEARNER PROFILE LINK

INQUIRERS Tracing how sight works step by step is real inquiry.
시각의 단계를 따라가 봄 — Inquirer.

KNOWLEDGEABLE Naming the eye's parts and their jobs builds knowledge.
눈의 부분과 역할을 앎 — Knowledgeable.

KOREAN BRIDGE

표현 "We see most things because light reflects off them." → "대부분의 물체는 빛이 반사되어 보인다."
기능(Function) 핵심 문장.

단어 "luminous / 발광하는 — reflect / 반사하다 — pupil / 동공 — retina / 망막"
핵심 단어 영-한.

Q2 Perspective DOK 3

In **"Animal Senses,"** trace THREE ways animals use their senses, then explain what "nature is the great compensator" means.

MODEL ANSWER

Sense 1 · Sight or Hearing: Many animals use sharp sight or hearing to find food and prey. A predator can spot or hear a small animal moving and then chase it, while a prey animal can use the same senses to notice danger early. **Sense 2 · Smell:** Animals use their sense of smell to find food and to protect themselves from predators. A familiar or strange smell can warn an animal that a hunter is near, or lead it toward something good to eat. **Sense 3 · Touch:** Animals use touch to find their way around their habitats, feeling their surroundings so they can move safely even where it is dark or crowded. **"Nature is the great compensator":** This phrase means that nature makes up for a sense that is weak by making another sense stronger, or by making the weak sense unnecessary. The narrator gives the rhinoceros as an example: a rhinoceros does not have good eyesight, but that is not a problem, because the rhino has no natural predators, so it does not need sharp eyes to stay safe. From an animal's perspective, each creature depends on the senses that matter most for its own way of life. That is how animals use their senses to navigate the world.

KEYWORDS (3)

sight/hearing = find food & prey | smell = stay safe from predators | compensator = a strong sense makes up for a weak one

SCORING NOTES

- Full credit (3): three senses with a real use for each + a clear explanation of "nature is the great compensator" using the rhinoceros.
- Partial (2): three senses named but the "compensator" idea is vague.
- Minimal (1): only "animals use their senses" with no example.

TEACHING STEPS

Step 1 · Three sense boxes.

강사: *"For sight/hearing, smell, and touch, write one way an animal uses it."*

코멘트. 세 칸 — 시각·청각, 후각, 촉각 각각 한 가지 쓰임 적기.

Step 2 · The rhino clue.

강사: *"Underline why weak eyesight is OK for the rhinoceros."*

코멘트. 코뿔소의 약한 시력이 괜찮은 이유에 밑줄.

Step 3 · Compensator sentence.

강사: *"Finish: 'Nature compensates by ____.'"*

코멘트. 문장 완성 — '자연은 ____함으로써 보충한다.'

STUDENT PROMPTS

- "Why might a prey animal need sharper hearing than a predator?"
  먹이 동물은 왜 포식자보다 더 예민한 청각이 필요할까?
- "How does touch help an animal in the dark?"

촉각은 어두운 곳에서 동물에게 어떻게 도움이 되니?

- "What sense would YOU most want to be stronger? Why?"
  너라면 어떤 감각이 더 강해지길 바라니? 왜?

DIFFERENTIATION

STRUGGLING Match 3 cards (find food / stay safe / find the way) to a sense.
카드 3장 — 먹이 찾기·안전·길 찾기를 감각과 매칭.

ADVANCED Compare an animal's strongest sense to a human's strongest sense.
동물의 가장 강한 감각과 사람의 가장 강한 감각 비교.

COMMON MISCONCEPTIONS

PITFALL "All animals have the same senses." Each animal relies on different senses for its way of life.
모든 동물의 감각이 같지 않음 — 저마다 다른 감각에 의존.

PITFALL "Weak eyesight always means danger." For the rhino it does not, because it has no natural predators.
약한 시력이 늘 위험은 아님 — 천적 없는 코뿔소엔 문제 아님.

LEARNER PROFILE LINK

OPEN-MINDED Seeing the world from an animal's sensory perspective is open-minded thinking.
동물의 감각 관점으로 봄 — Open-minded.

THINKERS Explaining "the great compensator" takes real reasoning.
'great compensator' 설명은 추론력 — Thinker.

KOREAN BRIDGE

표현 "Nature makes up for a weak sense with a stronger one." → "자연은 약한 감각을 더 강한 감각으로 보충한다."
관점(Perspective) 핵심 문장.

단어 "predator / 포식자 — prey / 먹이 — sense / 감각 — habitat / 서식지"
핵심 단어 영-한.

Q3 Responsibility DOK 3

In **"Blind Ambition,"** show what Matthew cannot do with sight, and how he uses other senses and help to do it anyway — and what "being blind makes me a better golfer" shows.

MODEL ANSWER

The Challenge: Matthew is blind. He cannot see the streets he crosses, the hallways at school, the friends he plays paintball with, the controls for the school play's sound system, or the golf hole he is aiming at. Sight cannot help him with any of these. **How Matthew Adapts:** Instead of denying his situation, Matthew has accepted it and takes responsibility for adapting. He depends on his other senses and on help from others. His guide dog, Twyla, uses her training to help him cross streets, get around obstacles, and move through crowded hallways. To play paintball, Matthew has his friends wear buzzers so he can hear where they are. To run the sound system, he puts Braille labels on the controls so he can feel which is which. To play golf, his coach or dad positions his club behind the ball and tells him how far the hole is, and then he swings. **What his words show:** When Matthew says, "I think being blind makes me a better golfer," he shows that he chooses to see his life in a hopeful way. He believes "there's always something life takes from you, and something life gives back." Rather than feeling sorry for himself, he focuses on what he CAN do. This is how he uses his senses — and his attitude — to navigate the world.

KEYWORDS (3)

challenge = cannot rely on sight

adapt = buzzers, Braille labels, Twyla, coach

attitude = life takes and gives back

SCORING NOTES

- Full credit (3): what sight cannot do + at least two real ways Matthew adapts + what his "better golfer" line shows about his attitude.
- Partial (2): names adaptations but no link to his attitude / accepting and adapting.
- Minimal (1): only "he has a guide dog" with no other detail.

TEACHING STEPS

Step 1 · Two columns.

강사: *"Make two columns: The Challenge / How Matthew Adapts. Fill in both."*

코멘트. 두 칸 — 어려움 / 매튜의 적응법. 둘 다 채우기.

Step 2 · The senses used.

강사: *"Circle which sense each adaptation uses (hearing buzzers, feeling Braille)."*

코멘트. 각 적응법이 쓰는 감각에 동그라미(버저=청각, 점자=촉각).

Step 3 · The attitude.

강사: *"Finish: 'Matthew chooses to focus on ___ instead of ___.'"*

코멘트. 문장 완성 — '매튜는 ____ 대신 ____에 집중하기로 한다.'

STUDENT PROMPTS

- "Why does Matthew put Braille labels on the sound controls?"
  매튜는 왜 음향 조절기에 점자 라벨을 붙일까?

- "How is trusting Twyla a kind of teamwork?"
  트와일라를 믿는 것은 어떻게 팀워크가 되니?
- "When have YOU adapted to do something a new way?"
  너는 무언가를 새로운 방법으로 해낸 적이 있니?

DIFFERENTIATION

STRUGGLING Match 3 cards (buzzers / Braille labels / Twyla) to what they help Matthew do.
카드 3장 — 버저·점자 라벨·트와일라를 도움 내용과 매칭.

ADVANCED Explain how Matthew "accepts" and "adapts" using both vocabulary words.
'accept'와 'adapt'를 모두 써서 매튜를 설명.

COMMON MISCONCEPTIONS

PITFALL "Twyla does everything for Matthew." Matthew himself adapts; Twyla helps, but he uses his own senses and ideas too.
트와일라가 다 해 주는 게 아님 — 매튜 스스로 적응하고 감각을 씀.

PITFALL "Being blind makes life sad for Matthew." He focuses on what he can do and even finds an upside.
매튜의 삶이 슬프기만 한 게 아님 — 할 수 있는 일에 집중하고 좋은 면도 찾음.

LEARNER PROFILE LINK

RISK-TAKERS Trying paintball and golf in new ways takes courage.
페인트볼·골프를 새 방식으로 시도 — Risk-taker.

BALANCED Matthew sees both what life takes and what it gives back.
삶이 가져가는 것과 주는 것을 함께 봄 — Balanced.

KOREAN BRIDGE

표현 "He uses his other senses to do what he loves." → "그는 다른 감각을 사용해 좋아하는 일을 해낸다."
책임(Responsibility) 핵심 문장.

단어 "accept / 받아들이다 — adapt / 적응하다 — obstacle / 장애물 — comfort / 위로하다"
핵심 단어 영-한.

Q4 Causation DOK 3

In **"The Game of Silence,"** compare what the game is like at the START with what it MEANS by the END, and explain how silence helps the children understand serious news.

MODEL ANSWER

The Game at the Start: At first the game of silence is just a children's game and a battle of wills. Omakayas is sure she will win. Her annoying younger brother, Pinch, keeps trying to make her laugh or cry out — he steals her doll, makes faces, and slips a burning wand next to her ankle. To Omakayas the rule of staying silent almost feels absurd, and the whole thing is playful and a little funny. **What Silence Means by the End:** By the end, the silence has become serious. The children must stay quiet so the adults can talk about what is happening to their people — a neighboring tribe has been forced from its land. That night, for the first time, everybody got their prizes and nobody lost, "for that night they knew the threat of a much bigger loss... That something was home." The game has turned into a way of understanding how grave the moment is. **How silence helps:** Being silent forces the children to stop playing and start paying close attention. When they listen instead of talk, they can hear and feel the seriousness of the adults' meeting. Their quiet, careful attention helps them realize that the biggest loss of all would be the loss of their home. This is causation: the enforced silence CAUSES the children to listen, and listening leads them to understanding.

KEYWORDS (3)

start = playful game, Pinch teases

end = serious; the bigger loss is home

silence → attention → understanding

SCORING NOTES

- Full credit (3): the playful start + the serious end (loss of home) + how silence/attention leads to understanding.
- Partial (2): start and end named but no link from silence to understanding.
- Minimal (1): only "they play a quiet game" with no change in meaning.

TEACHING STEPS

Step 1 · Start vs. end.

강사: *"Make two columns: the game at the START / what silence MEANS by the END."*

코멘트. 두 칸 — 시작의 게임 / 끝에서 침묵의 의미.

Step 2 · The big loss.

강사: *"Circle the words that name the 'much bigger loss' (home)."*

코멘트. 'much bigger loss'를 가리키는 말(home)에 동그라미.

Step 3 · Silence sentence.

강사: *"Finish: 'Staying silent helps the children ____.'"*

코멘트. 문장 완성 — '침묵을 지키는 것은 아이들이 ____하도록 돕는다.'

STUDENT PROMPTS

- "Why does Pinch try so hard to make Omakayas talk?"
  핀치는 왜 그렇게 오마카야스를 말하게 하려 애쓸까?
- "How can being quiet help you notice more?"

조용히 있으면 어떻게 더 많이 알아차릴 수 있니?

- "When did staying quiet help YOU understand something?"
  조용히 있어서 무언가를 더 잘 이해한 적은 언제니?

DIFFERENTIATION

STRUGGLING Match 2 cards ("just a game" / "serious news about home") to start and end.

카드 2장 — '그냥 게임'과 '집에 관한 심각한 소식'을 시작·끝과 매칭.

ADVANCED Explain how the story uses Ojibwe words and similes to set the scene in 1850.

이야기가 오지브웨 단어와 직유로 1850년 배경을 그리는 방식 설명.

COMMON MISCONCEPTIONS

PITFALL "The story is only about a brother and sister fighting." The game turns into a way to grasp serious news about losing their home.

남매 다툼 이야기만은 아님 — 집을 잃을 심각한 소식을 깨닫는 과정.

PITFALL "Pinch and Two Strike are the same person." Pinch is Omakayas's brother; Two Strike and the Angry One are separate cousins/visitors.

핀치와 투 스트라이크는 다른 인물 — 핀치는 남동생, 투 스트라이크·앵그리 원은 사촌/손님.

LEARNER PROFILE LINK

REFLECTIVE Noticing how a game becomes serious is reflective thinking.

게임이 진지해지는 흐름을 알아챔 — Reflective.

CARING The children grow to care about their family's home and safety.

가족의 집과 안전을 염려하게 됨 — Caring.

KOREAN BRIDGE

표현 "Silence and attention reveal what matters most." → "침묵과 주의는 가장 중요한 것을 드러낸다."

인과(Causation) 핵심 문장.

단어 "silence / 침묵 — forfeit / 몰수하다 — taunt / 조롱 — home / 집·고향"

핵심 단어 영-한.

Q5 Connection DOK 4

Essential Question: How do people and animals use their senses to navigate the world? Use ideas from TWO of the selections in this module to answer.

MODEL ANSWER

People and animals use their senses to gather information about the world and to stay safe, find what they need, and understand what is happening around them. In **"The Science Behind Sight,"** we learn how the sense of sight works. Light reflects off objects and enters the eye through the pupil and lens, lands on the retina, and travels through the optic nerve to the brain, which makes sense of the picture. Animals like foxes and cats have eyes built for their needs, so they can hunt or see in the dark. Sight lets people and animals know the size, shape, color, and distance of things around them. In **"Blind Ambition,"** Matthew cannot use sight at all, yet he still navigates the world by leaning on his other senses. He listens for buzzers to play paintball, feels Braille labels to run a sound system, and trusts his guide dog Twyla to lead him safely. **My Big Idea:** Senses are how every living thing connects to the world. As Immanuel Kant said, "All our knowledge begins with the senses." When one sense is missing or weak, people and animals adapt by using their other senses and by helping one another, so they can still find their way through the world.

KEYWORDS (3)

Science Behind Sight = how sight gathers information

Blind Ambition = other senses replace sight

Big Idea: senses connect us to the world

SCORING NOTES

- Full credit (4): two selections named + which senses each shows + how they help navigate the world + a Big Idea answering the EQ.
- Partial (3): two selections + senses + Big Idea weak.
- Developing (2): one selection only.
- Minimal (1): generic ("senses are important") without naming a selection.

SYNTHESIS FRAME

SELECTION 1 IDEA

Name the senses and how they help: The Science Behind Sight = how sight works; Animal Senses = animals use sight, smell, hearing, touch; Blind Ambition = other senses replace sight; The Game of Silence = silence sharpens listening and attention.

CONNECTING WORD

BOTH / AND — joins two selections (for example Sight AND Blind Ambition) into one Big Idea about how senses help us navigate the world.

SELECTION 2 IDEA

A different selection than Selection 1 — so the two show two ways the senses help a person or animal find its way.

REAL-WORLD CONNECTION

KOREA 우리도 감각으로 세상을 읽는다 — 길을 건널 때 차 소리를 듣고, 음식이 상했는지 냄새로 알고, 어두운 방에서 손으로 더듬어 길을 찾는다. 한 감각이 약해지면 다른 감각이 그 자리를 메운다.

내가 매일 쓰는 감각 하나를 떠올려 EQ와 연결하기.

PERSONAL Which sense do you rely on most, and how would you adapt if you could not use it?

네가 가장 많이 의존하는 감각은 무엇이고, 그것을 못 쓰면 어떻게 적응하겠니?

RUBRIC

LEVEL	WHAT IT LOOKS LIKE
Exceeding (4)	Names 2 selections + which senses each shows + how they help navigate the world + a Big Idea answering the EQ.
Meeting (3)	Names 2 selections + a sense for each + a Big Idea.
Approaching (2)	Names 2 selections but only retells facts — no link to navigating the world.
Beginning (1)	Generic ("senses matter") without text evidence.

Watch out: Many G4 students will just list facts (the eye has a retina; Matthew has a guide dog) and stop. Push them to connect each SENSE to HOW it helps a person or animal navigate the world, and to write a Big Idea. The full EQ asks HOW senses help people and animals find their way, not just what the senses are.

TEACHING STEPS

Step 1 · Two selection cards.

강사: *"Pick 2 selections. For each, name a sense and who uses it."*

코멘트. 선택지 둘 — 각자 감각 하나와 사용하는 주체 적기.

Step 2 · The how.

강사: *"Under each, write how that sense helps them navigate the world."*

코멘트. 각각 아래 — 그 감각이 세상을 헤쳐 가는 데 어떻게 돕는지.

Step 3 · Big Idea sentence.

강사: *"Write one sentence answering the EQ. Use both selections."*

코멘트. EQ에 답하는 한 문장 — 두 선택지 모두 사용.

STUDENT PROMPTS

- "Which sense matters most in each selection you chose?"
  고른 선택지마다 가장 중요한 감각은 무엇이니?
- "How do living things adapt when one sense is missing?"
  한 감각이 없을 때 생명체는 어떻게 적응하니?
- "Why does Kant say knowledge begins with the senses?"
  칸트는 왜 지식이 감각에서 시작된다고 말했을까?

DIFFERENTIATION

STRUGGLING Sentence frame: "In ____, ____ uses the sense of ____ to ____. Senses help us navigate the world because ____."
문장 골격 제공.

ADVANCED Use THREE selections and sort them by which sense each highlights (sight / smell / touch / hearing).

선택지 셋 사용 — 강조하는 감각별 분류(시각·후각·촉각·청각).

COMMON MISCONCEPTIONS

PITFALL "Only sight matters." Smell, hearing, touch, and taste all help people and animals navigate the world.

시각만 중요한 게 아님 — 후각·청각·촉각·미각도 세상을 헤쳐 가게 도움.

PITFALL "Listing facts answers the EQ." The EQ asks HOW senses help us navigate, not just what they are.

사실 나열은 EQ 답이 아님 — '감각이 어떻게 길을 찾게 돕는지'를 물음.

LEARNER PROFILE LINK

THINKERS Building one Big Idea from two selections is the heart of critical thinking.

두 선택지로 하나의 Big Idea를 세움 — Thinker.

COMMUNICATORS Explaining how senses connect us to the world is clear communication.

감각이 세상과 우리를 잇는 방식을 설명 — Communicator.

KOREAN BRIDGE

표현 "Our senses connect us to the world." → "우리의 감각이 우리를 세상과 이어 준다."

M2 마무리 문장 — 학생 글쓰기 핵심.

인용 "All our knowledge begins with the senses." — Immanuel Kant → "우리의 모든 지식은 감각에서 시작된다."

에피그래프 — 감각과 앎.

Worksheet D — MAP Practice · Answer Key

20 items · RIT 200 → 220 · Lexile 500L–800L

MAP Mini-Passage Word Counts (Worksheet D)

- Passage 1 · The Owl's Night Eyes · **167 words** · Lexile 500–560L · RIT 200–206
- Passage 2 · How Dogs Smell the World · **172 words** · Lexile 560–620L · RIT 206–211
- Passage 3 · Reading with Your Fingers · **172 words** · Lexile 620–700L · RIT 211–216
- Passage 4 · The Taste Test · **184 words** · Lexile 700–800L · RIT 216–220

All four passages within target band 150–220 words; RIT 200→220; Lexile 500L–800L. Word counts match the workbook passage meta exactly. Each passage models a different way the senses help navigate the world — an owl's night vision and hearing, a dog's sense of smell, the sense of touch through braille, and how taste and smell work together. Passages are NEW — no recall of the four anchor selections.

PASSAGE 1 · THE OWL'S NIGHT EYES

Q1 RIT 200

A

Lexile 500L · Key Detail. "an owl glides through the dark forest searching for food." *Distractors B/C/D say day / noon / storm.*

Q2 RIT 202

B

Lexile 520L · Cause & Effect. "Big eyes can take in more light, so even a little moonlight is enough." *Trap A/C/D off-text.*

Q3 RIT 203

C

Lexile 530L · Key Detail. "an owl's eyes cannot move from side to side. To look around, an owl must turn its whole head." *Trap A/B/D off-text.*

Q4 RIT 205

D

Lexile 550L · Inference. Finding a mouse "even when the owl cannot see it" shows it relies on hearing when its eyes are not enough. *Trap A/B/C miss it.*

Q5 RIT 206

A

Lexile 560L · Main Idea. "Its body is built perfectly for a life spent awake at night." *Trap B/C/D narrow or off-topic.*

PASSAGE 2 · HOW DOGS SMELL THE WORLD

Q6 RIT 206

B

Lexile 560L · Key Detail. "dogs learn about the world mostly by smelling it." *Trap A/C/D name other senses.*

Q7 RIT 208

C

Lexile 580L · Key Detail. "a dog can have more than two hundred million." *Trap A is the human number (about five million).*

Q8 RIT 209

D

Lexile 590L · Cause & Effect. a dog "can notice smells that are much too faint for a person to detect" because it has far more smell sensors. *Trap A/B/C off-text.*

Q9 RIT 210

A

Lexile 600L · Inference. Smelling one spot a long time, the dog is "reading the world" — gathering information. *Trap B/C/D misread it.*

Q10 RIT 211

B

Lexile 620L · Main Idea. a dog reads the world "through its remarkable sense of smell." *Trap A/C/D narrow or off-topic.*

PASSAGE 3 · READING WITH YOUR FINGERS

Q11 RIT 211

C

Lexile 620L · Key Detail. "a French boy named Louis Braille changed everything." *Trap A/B/D off-text.*

Q12 RIT 212

A

Lexile 640L · Cause & Effect. "Louis had lost his sight... he wished he could read the way other students did." *Trap B/C/D off-text.*

Q13 RIT 213

B

Lexile 660L · Key Detail. "Each small pattern of dots stands for a letter, a number, or even a musical note." *Trap A/C/D off-text.*

Q14 RIT 215

B

Lexile 680L · Vocabulary. a "doorway to reading" = a way to reach or get to something. *Trap A/C/D take it literally.*

Q15 RIT 216

D

Lexile 700L · Main Idea. "Braille turns the sense of touch into a doorway to reading." *Trap A/B/C trivialize it.*

PASSAGE 4 · THE TASTE TEST

Q16 RIT 216

A

Lexile 700L · Key Detail. "He set a small cup of juice and a soft blindfold on each desk." *Trap B/C/D off-text.*

Q17 RIT 217

B

Lexile 720L · Cause & Effect. with nose clips on, "the juice tasted dull and plain," because they could not smell it. *Trap A/C/D off-text.*

Q18 RIT 218

C

Lexile 740L · Key Detail. "taste and smell are partners." *Trap A/B/D name the wrong pair.*

Q19 RIT 219

D

Lexile 770L · Inference. "food seems flavorless when you have a stuffy nose" — a blocked nose blocks smell, a big part of taste. *Trap A/B/C miss it.*

Q20 RIT 220

D

Lexile 800L · Main Idea. "our senses do not work alone. They team up... together they tell us far more than any single sense could." *Trap A/B/C trivialize it.*